The Fibonacci-distance Neighborhood for the Permutation Flow Shop Problem

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Introduction

- **Permutation Flow Shop Problem (PFSP):** NP-hard for $m \geq 3$. Neighborhood choice drives metaheuristic quality.
- Classical neighborhoods trade off coverage and cost:
  - **Adjacent swap:** $n-1$ moves, only short-range.
  - **Dynasearch:** $O(n^2)$ candidates, expensive per call — intractable beyond $n \sim 200$.
- **Fibonacci-distance insertion:** $O(n \log n)$ moves spanning short *and* long range. Stays tractable up to $n = 500$ and beyond.
- This talk: define the Fibonacci neighborhood, study its scaling behavior on Taillard benchmarks, and discuss its role within local search frameworks (ILS, SA).

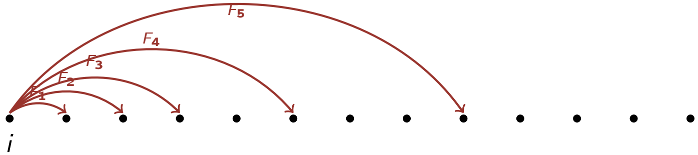
The Fibonacci-distance Neighborhood

Definition. Given $\pi = (\pi_0, \ldots, \pi_{n-1})$, define moves

$$N_{\mathrm{fib}}(\pi) \;=\; \{\, \mathrm{insert}(\pi_i \to i + F_k) \;:\; 0 \le i < n,\ F_k \le n - i \,\},$$

where $F_k = 1, 2, 3, 5, 8, 13, 21, \ldots$ are Fibonacci numbers.
Size: $|N_{\mathrm{fib}}(\pi)| = O(n \log n)$. Local + long-range coverage from a single rule.

Idea. A geometric, golden-ratio-scaled set of move distances from position i:



Why Fibonacci? — Properties

- **Golden-ratio scaling.** Consecutive jump distances grow by $\varphi \approx 1.618$ — the densest non-uniform integer sequence avoiding any short cycle.
- **Logarithmic per-position.** Only $O(\log n)$ jump lengths per starting position; total $O(n \log n)$ moves *vs.* $O(n^2)$ for full insertion or dynasearch.
- **Tridiagonal QUBO.** The QUBO matrix has bandwidth 2 — embeds natively on D-Wave with no chain breaks. Scales cleanly to $n = 500$.
- **Self-similarity.** Fibonacci shifts within a Fibonacci shift form a sub-neighborhood — the move structure is recursive, well-suited to multi-scale local search.

QUBO Formulation for D-Wave

The Fibonacci neighborhood admits a **tridiagonal QUBO** of bandwidth 2:

$$Q_{\mathrm{fib}}(x) = \sum_{i=0}^{n-1} a_i x_i + \sum_{i=0}^{n-2} b_i x_i x_{i+1},$$

where $x_i \in \{0, 1\}$ selects the Fibonacci jump at position i and the b_i couplers encode swap conflicts between consecutive positions.

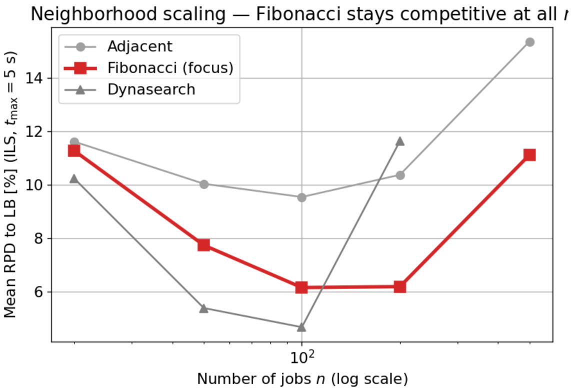
Why this matters on D-Wave Advantage:

- Bandwidth 2 $\Rightarrow$ trivial linear-chain embedding on Pegasus/Zephyr, no chain breaks.
- Number of logical qubits $= n$ (one per position), feasible up to $n \leq 500$.
- Acceleration by the Smutnicki block-property and the NPI (no-positive-improvement) filter further prunes the candidate set before submission.

Experimental Setup

- **Benchmarks:** Taillard instances, $n \in \{20, 50, 100, 200, 500\}$, $m \in \{5, 10, 20\}$, 10 instances each, 5 seeds.
- **Time limits:** $t_{\max} \in \{0.1, 0.5, 1, 2, 5, 10\}$ s; metric: mean RPD to lower bound.
- **Algorithms:** Iterated Local Search (tabu τ=10, MushroomList k=10); Simulated Annealing (T_0=1000, α=0.99, reheat-kick on stagnation).
- **Quantum:** D-Wave Advantage 4.1, 5,000 reads, windowed decomposition for $n > 19$.
- **Compared neighborhoods:** Adjacent, **Fibonacci**, Dynasearch.

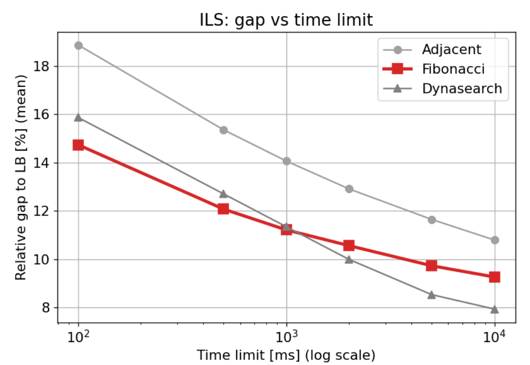
Scaling Behavior — RPD vs. n

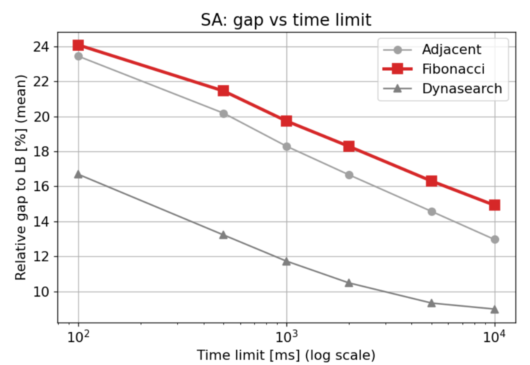


Observations (ILS, $t_{\max} = 5\,\mathrm{s}$, m=10):

- Up to $n \approx 100$: Dynasearch wins, Fibonacci second.
- $n = 200$: Fibonacci **overtakes** Dynasearch (6.19 % vs. 11.63 %).
- $n = 500$: Dynasearch becomes intractable; Fibonacci continues at **11.12 %** — the only classical neighborhood that still scales.

Gap vs. Time Limit





- Fibonacci has the flattest curve — already near its plateau at $t_{\max} = 100$ ms.
- Dynasearch reaches lower RPD at small n but needs the full 10 s budget to get there.

Conclusion and Future Work

Summary:

- Fibonacci is the only classical elementary neighborhood that stays tractable from $n = 20$ to $n = 500$.
- Tridiagonal QUBO structure embeds cleanly on D-Wave Advantage — no chain breaks, one qubit per position.
- Empirically beats Dynasearch from $n \geq 200$, where Dynasearch can only complete a single DP call within the time budget.

Future work:

- Full QPU evaluation of the Fibonacci QUBO at $n = 500$ on D-Wave Advantage.
- Theoretical analysis of the golden-ratio jump distribution vs. uniform / geometric alternatives.
- Fibonacci-of-Fibonacci composite moves — recursive self-similar generalization.